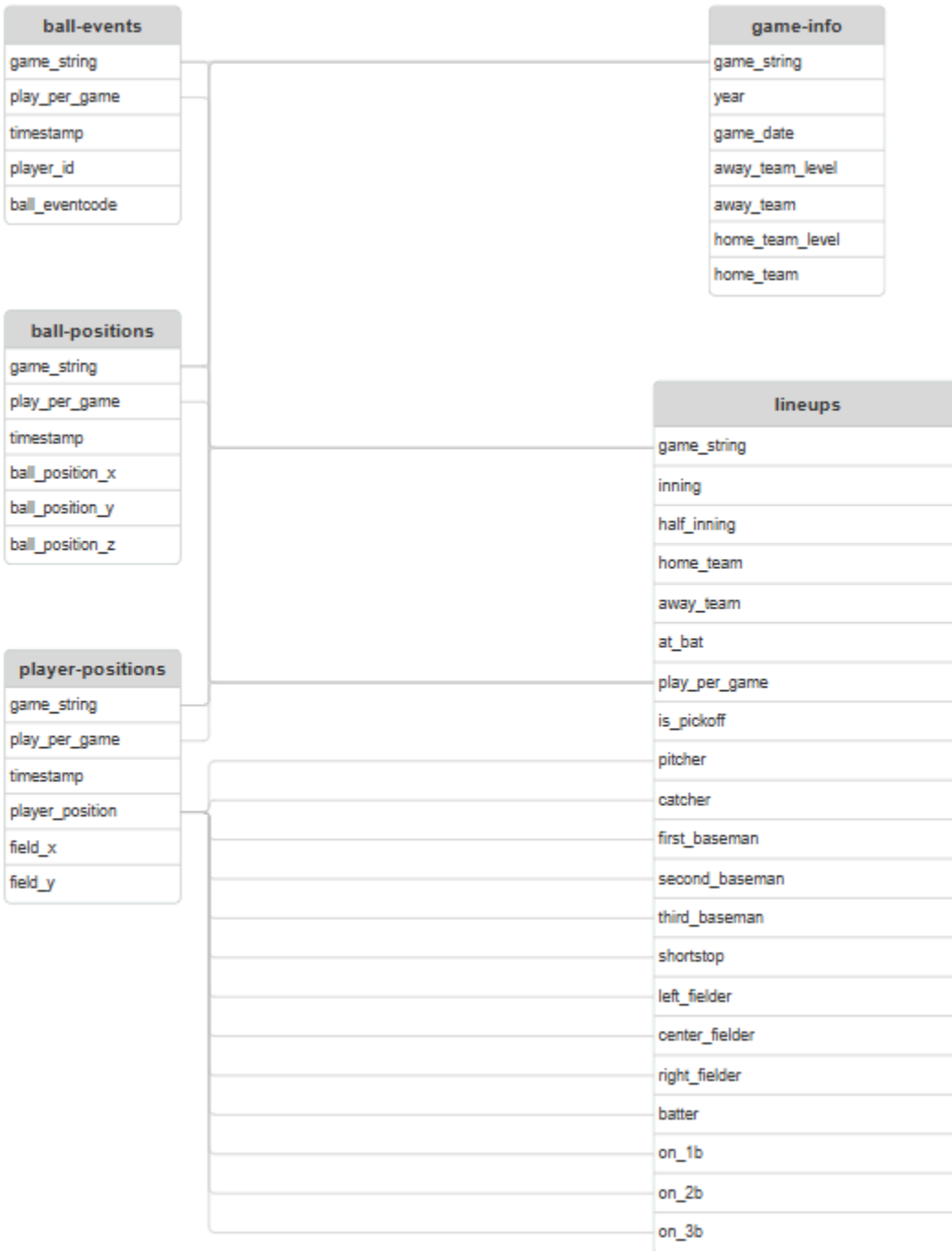


I. Entity Relationship Diagram



II. Data Dictionary

Table: ball-events

- **game_string** - Fields are structured “y{year}_d{day}_ {away}_ {home}”. Each game string identifies the year and the sequential day within the season.
- **play_per_game** - A play is defined as a situation where the ball is live. Each play_per_game generally include at least three events (identified by ball_eventcode): the pitch, what happens to the ball, and the end of the play. play_per_game begin at 1 and are listed consecutively for each game.
- **timestamp** - Times are shown in milliseconds, 1 sec = 1000 ms. Timestamps for a given play_per_game begin 3 seconds before the ball is released
- **player_id** - Corresponding position numbers are given in the Glossary. NOTE: Each player position corresponds to a lineups column, and those column names are connected through the Glossary.
- **ball_eventcode** - Numbers corresponding to ball events are given in the Glossary.

Table: ball-positions

- **game_string** – See description in ball-events
- **play_per_game** - See description in ball-events
- **timestamp** – See description in ball-events
- **ball_position_x** – $x = 0$ is defined at the line between home plate and second base, with $x > 0$ towards first base. Dimensions are given in feet.
- **ball_position_y** - $y = 0$ is at the back of home plate, with $y > 0$ towards second base. Dimensions are given in feet.
- **ball_position_z** - z is the height with respect to $z=0$ in real-world coordinates. $z=0$ does not necessarily correspond to the ground, which is neither flat nor level. Dimensions are given in feet

Table: game-info

- **game_string** – See description in ball-events
- **year** – Anonymized Year that the game took place
- **game_date** – Anonymized date that the game took place. Dates are labeled sequentially from the beginning of the season
- **away_team_level** – Anonymized Minor League Baseball level for the away team. Levels are labeled sequentially; for example, level 1 is lower than level 2.
- **away_team** – Random 3-letter team code representing the away team
- **home_team_level** – Anonymized Minor League Baseball level for the home team. Levels are labeled sequentially; for example, level 1 is lower than level 2.
- **home_team** – Random 3-letter team code representing the home team

Table: player-positioning

- **game_string** – See description in ball-events
- **play_per_game** - See description in ball-events
- **timestamp** – See description in ball-events
- **player_id** – See description in ball-events
- **field_x** – Same coordinates and dimensions as ball_position_x in ball-positions table
- **field_y** - Same coordinates and dimensions as ball_position_y in ball-positions table

Table: lineups

- **game_string** – See description in ball-events
- **inning** – Calculated column that states what inning the game is in.
- **half_inning** – Calculated column that states if the game is in the Top or Bottom of the Inning. The away team bats in the top of the inning. The home team bats in the bottom.
- **home_team** – See description in game-info
- **away_team** – See description in game-info
- **at_bat** – Calculated column that corresponds to a batter at the plate. They are listed consecutively for each game and includes at least one play_per_game.
- **play_per_game** – See description in ball-events
- **is_pickoff** – Boolean column that states if the play_per_game is a pickoff
- **pitcher** – Populated by a player code in the form: "{3 Letter Team Code}-####". Corresponds with player_id 1 in the player-positioning and ball-events tables
- **catcher** – Populated by a player code in the form: "{3 Letter Team Code}-####". Corresponds with player_id 2 in the player-positioning and ball-events tables
- **first_base** – Populated by a player code in the form: "{3 Letter Team Code}-####". Corresponds with player_id 3 in the player-positioning and ball-events tables
- **second_base** – Populated by a player code in the form: "{3 Letter Team Code}-####". Corresponds with player_id 4 in the player-positioning and ball-events tables
- **third_base** – Populated by a player code in the form: "{3 Letter Team Code}-####". Corresponds with player_id 5 in the player-positioning and ball-events tables
- **shortstop** – Populated by a player code in the form: "{3 Letter Team Code}-####". Corresponds with player_id 6 in the player-positioning and ball-events tables
- **left_fielder** – Populated by a player code in the form: "{3 Letter Team Code}-####". Corresponds with player_id 7 in the player-positioning and ball-events tables
- **center_fielder** – Populated by a player code in the form: "{3 Letter Team Code}-####". Corresponds with player_id 8 in the player-positioning and ball-events tables

- **right_fielder** – Populated by a player code in the form: "{3 Letter Team Code}-####". Corresponds with player_id 9 in the player-positioning and ball-events tables
- **batter** – Populated by a player code in the form: "{3 Letter Team Code}-####". Corresponds with player_id 10 in the player-positioning and ball-events tables
- **on_1b** – Populated by a player code in the form: "{3 Letter Team Code}-####". Corresponds with player_id 11 in the player-positioning and ball-events tables
- **on_2b** – Populated by a player code in the form: "{3 Letter Team Code}-####". Corresponds with player_id 12 in the player-positioning and ball-events tables
- **on_3b** – Populated by a player code in the form: "{3 Letter Team Code}-####". Corresponds with player_id 13 in the player-positioning and ball-events tables

III. Glossary

ball-events ball_eventcode Lookup	
ball_eventcode	Definition
1	Pitch
2	Ball Acquired
3	Throw (ball-in-play)
4	Ball Hit Into Play
5	End of Play
6	Pickoff Throw
7	Ball Acquired – Unknown Field Position
8	Throw (ball-in-play) – Unknown Field Position
9	Ball Deflection
10	Ball Deflection Off Wall
11	Home Run
16	Ball Bounce

ball-events/player-positioning player_id Lookup	
player_id	Definition
1	Pitcher
2	Catcher
3	First Baseman
4	Second Baseman
5	Third Baseman
6	Shortstop
7	Left Field
8	Center Field
9	Right Field
10	Batter
11	Runner on 1B
12	Runner on 2B
13	Runner on 3B
14	Home Plate Umpire
15-17	Field Umpire
18	First Base Coach
19	Third Base Coach
255	Ball event with no player (e.g., ball bounce)